Distributional Emulation

Robust & Rapid Emulation for Efficient Trial Design

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The problem

If $\hat{M}$ is slow (minutes-to-hours), it will take years-to-decades to compute p.

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What is an Emulator?

An emulator is a function $\hat{E}_M$ which *behaves* like $\hat{M}$, but which is vastly quicker to compute.

Estimators, Regression

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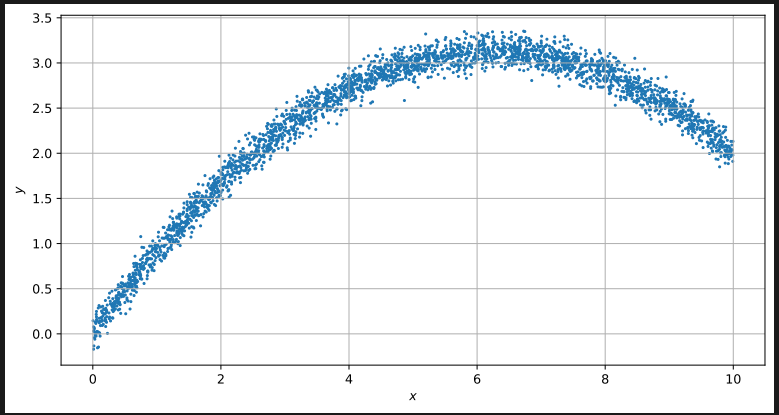

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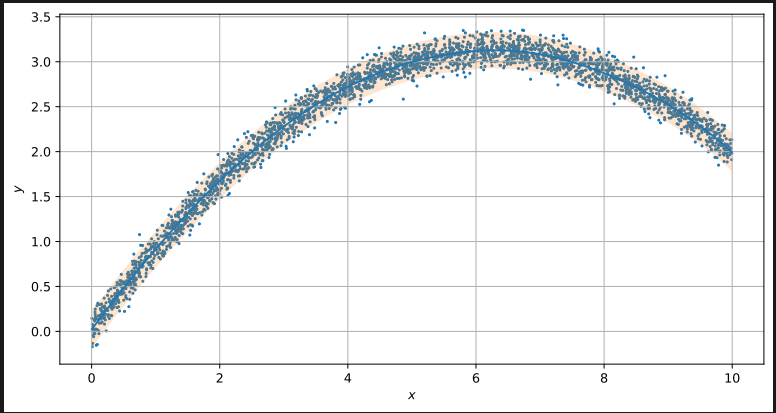

The Limitation

Distributions, not means

If $\hat{M}$ is stochastic, then $\boldsymbol{y}$ is not a number - it is a *realisation on a random process*. These methods serve only to predict the mean, $\mathbb{E}(\boldsymbol{y})$, not the distributions around that point; or 'paint' an assume distribution family on top of a point predictor.

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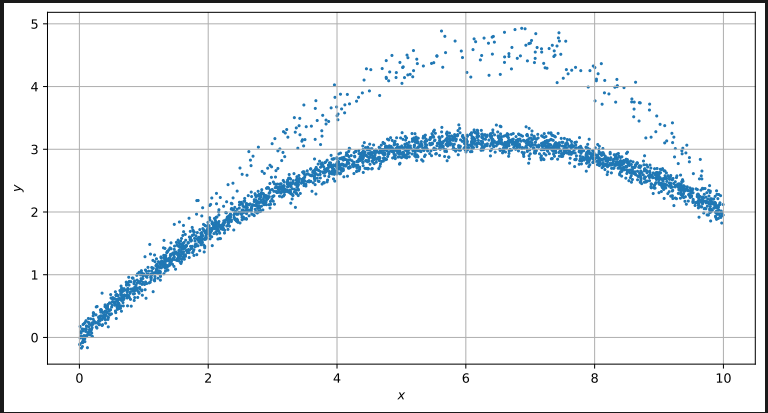

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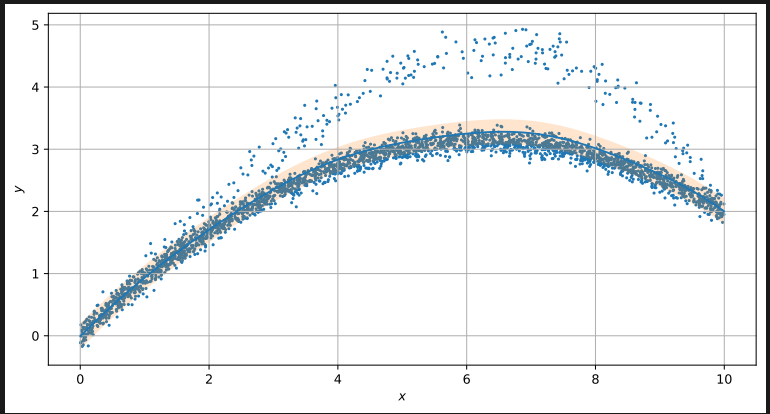

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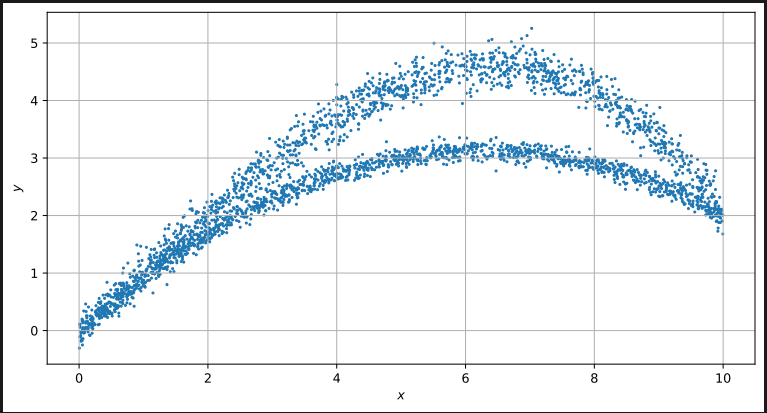

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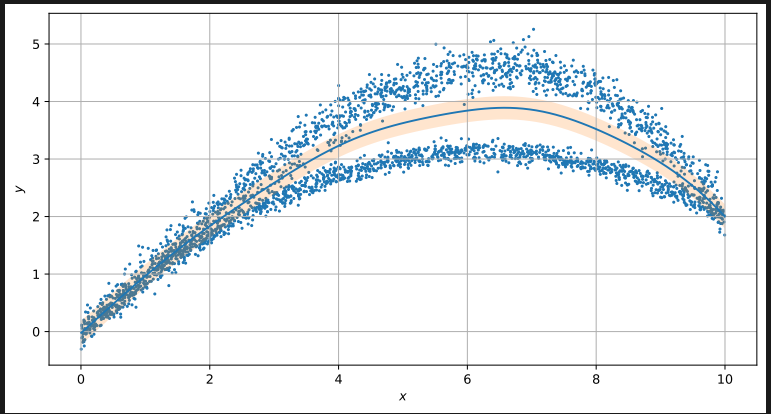

Machine Learning (Again)

There *are* ML models which give distributions (Bayesian Neural Networks), but we may choose to avoid them:

1. Require lots of data (expensive - recall computing $\hat{M}$ is the bottleneck)
2. Black boxes (aesthetically/scientifically dubious)

Gaussian Processes

A modelling process where one assumes

$$\boldsymbol{y}(\boldsymbol{\pi}) \sim \mathcal{N}\left(\sum_i w_i(\pi)\boldsymbol{y}_i, \Sigma\right) \tag{1}$$

Has many lovely properties, but fundamentally unsuitable for our work, as it assumes the distribution, rather than reconstructing it!

Partition Models

Split the covariate space up into *cells*

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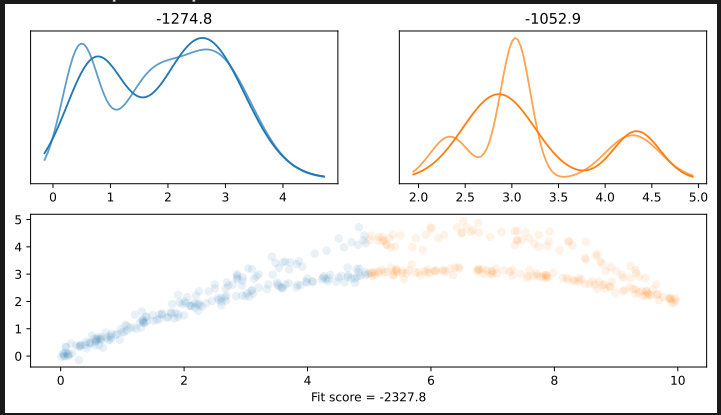

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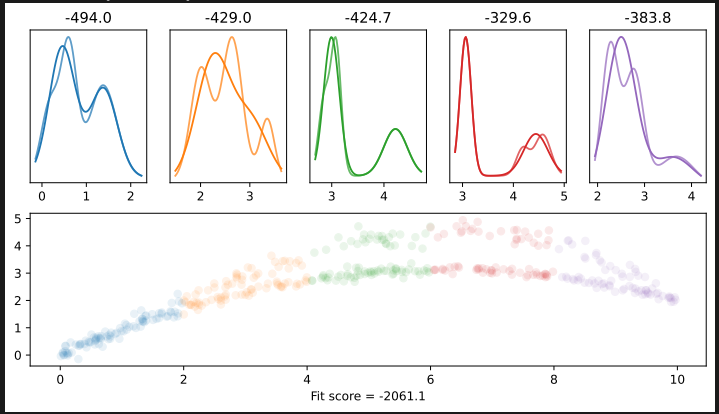

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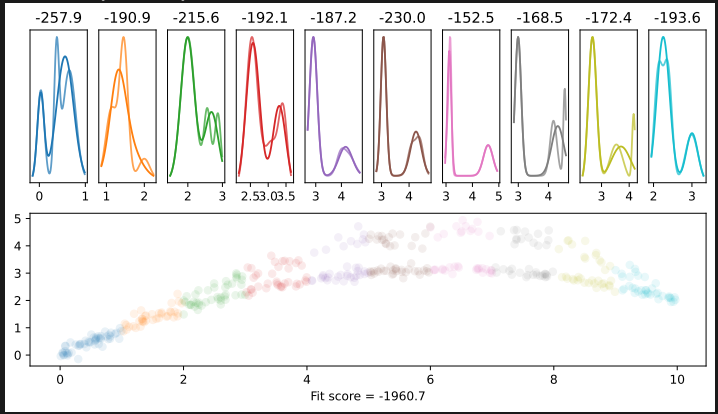

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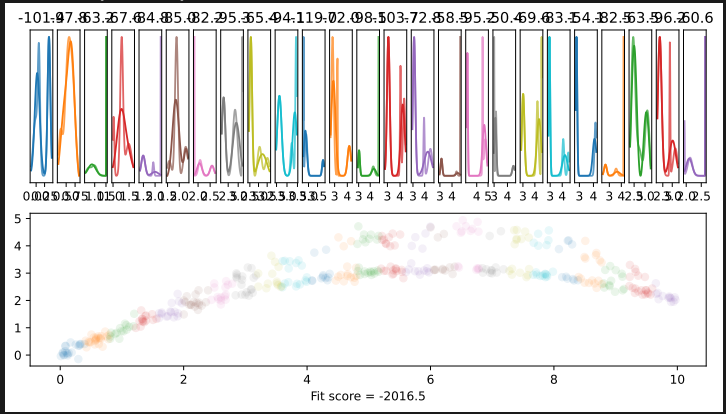
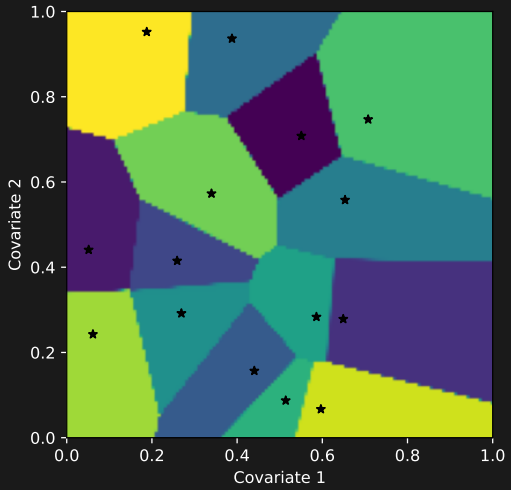


Partition Models

In higher dimensions, the partioning gets harder!



Mixture of Experts

Instead of a hard partition, MoE probabilistically assign the 'problem' to an 'expert':

$$p(y|\boldsymbol{x}) = \sum_i w_i(\boldsymbol{x}) p_i \tag{2}$$

The w_i give different weights to the 'expert opinion' at different points in covariate space, coming to a different consensus.
Standard MoE uses Neural Networks to determine w_i (via complex partitioning of a learned hyperplane projection)

Models are not Emulators

Important!

The above models *are not emulators*. They provide ways to model the best-fit posterior distribution, conditioned on the observed data:

$$\text{Model}_{\boldsymbol{\Theta}}(\boldsymbol{x}) = p(\boldsymbol{y}|\boldsymbol{x}, \boldsymbol{\Theta})$$

Posterior Predictive Distribution

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$$\text{Emulator}(\boldsymbol{x}) = p(\boldsymbol{y}|\boldsymbol{x}) = \int \text{Model}_{\boldsymbol{\Theta}}(\boldsymbol{x}) \mathrm{d}\boldsymbol{\Theta} \tag{3}$$

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1. Reconstruct probability distribution function
2. Allow both smooth and near-discontinuous changes in behaviour
3. Inherit 'expert knowledge' to reduce training requirements
4. Be interpretable and accessible

Department of Experts

A middle ground between partitioning and MoE models. We allow each expert to belong to a 'department' (a partition - which may have many experts within it), but also allow them to 'spend time' between departments.
The weight of an expert in the sum is then:

$$w_i(\boldsymbol{x}) = \sum_{\text{dep. } k} P(\boldsymbol{x} \in k) \times \frac{P(i \in k) P(i \text{ knows } \boldsymbol{x})}{\sum_j P(j \in k) P(j \text{ knows } \boldsymbol{x})} \tag{4}$$

These probabilities can be learned by the machine

The Clever Bit

This reduces trivially down into a standard MoE, with model parameters $\boldsymbol{\Theta}$

$$p(\boldsymbol{x}|\boldsymbol{\Theta}) = \sum_{w} w_i(\boldsymbol{x}|\Theta) p_i(\boldsymbol{x}|\boldsymbol{\Theta}) \tag{5}$$

The clever bit is that we allow each department to have their own 'grammar' that they use to talk to interpret the covariance space. The distance metric in each space is:

$$d_k(\boldsymbol{x}, \boldsymbol{y}) = \sqrt{(\boldsymbol{x} - \boldsymbol{y}) \cdot G_k(\boldsymbol{x} - \boldsymbol{y})} \tag{6}$$

Where G_k is a learned distance metric unique to each cell.

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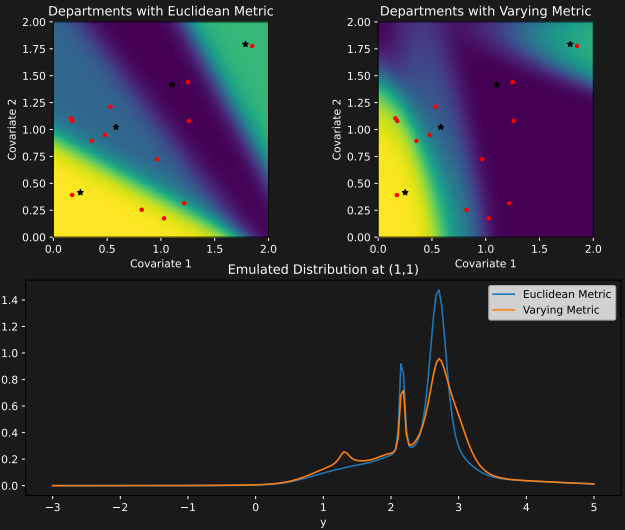
2. The metrics are easily interpreted as covariance measures localised within a department.
3. Gives the same predictive power as a high dimensional hidden-layer neural network
4. Constrained to produce 'interpretable results'.

A FADE Model



Project Ideas

As a group, we want to understand the capabilities (and limits) of our new tools: this is where you come in!

QUESTION ONE What is the best way to train a FADE instance on a given dataset? Are there more or less optimal starting positions? Splits between training and validation data?

QUESTION TWO What classes of models does FADE work best at emulating? What is it bad at? Can you show a transition point where FADE goes from good to bad, or vice versa?

QUESTION THREE Can FADE be applied to other, real world models, in other fields? How accurate is it? Does the FADE setup bring advantages that a more naive emulation might miss?

Getting Started

Important!

Don't try to get too advanced, too quickly: start small & build your way up

1. Familiarise yourself with the underlying theory of emulators: read the FADE paper
2. Generate a synthetic dataset of a 'model' that you have designed.
3. Try and train a FADE instance to emulate your model: how good is it? What changes make it better or worse?
4. Iterate and build on this, making the model more complex, and playing around with the FADE training and scoring.

Tools & Languages

This will be a computationally-heavy project, but there's no requirement to learn a language you aren't familiar with already.

- FADE itself is written in C++20.
  1. You don't have to modify this code or alter it unless you want to explore this route.
  2. You will, however, need to know how to set up config files for FADE, which will mean understanding what the options do.
- When generating synthetic datasets, you may use whatever language you like (the output must be in a standardised format, for FADE to read it).
- When exploring other 'real world' models, you'll need to ensure that you can actually run them